

GitHub Code Review

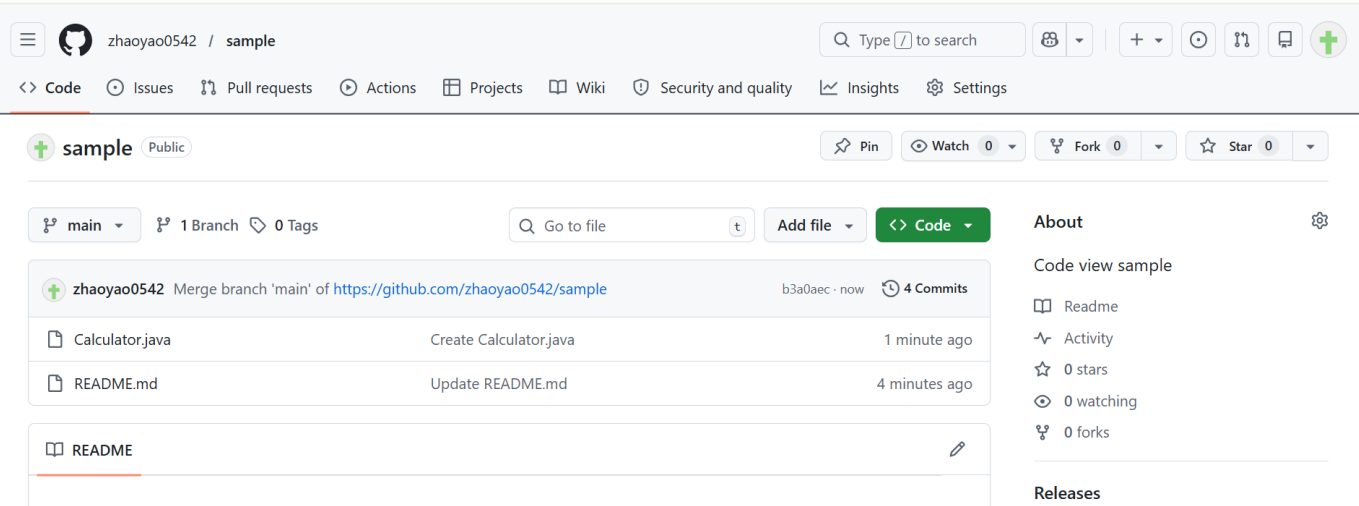
1. Pull Requests

Pull requests (PRs) are a core feature of GitHub that enable teams to review, discuss, and improve code before merging changes into the main branch. They are widely used for proposing new features, fixing bugs, and collaborating on implementation details.

 Reference: <https://github.com/features/code-review>

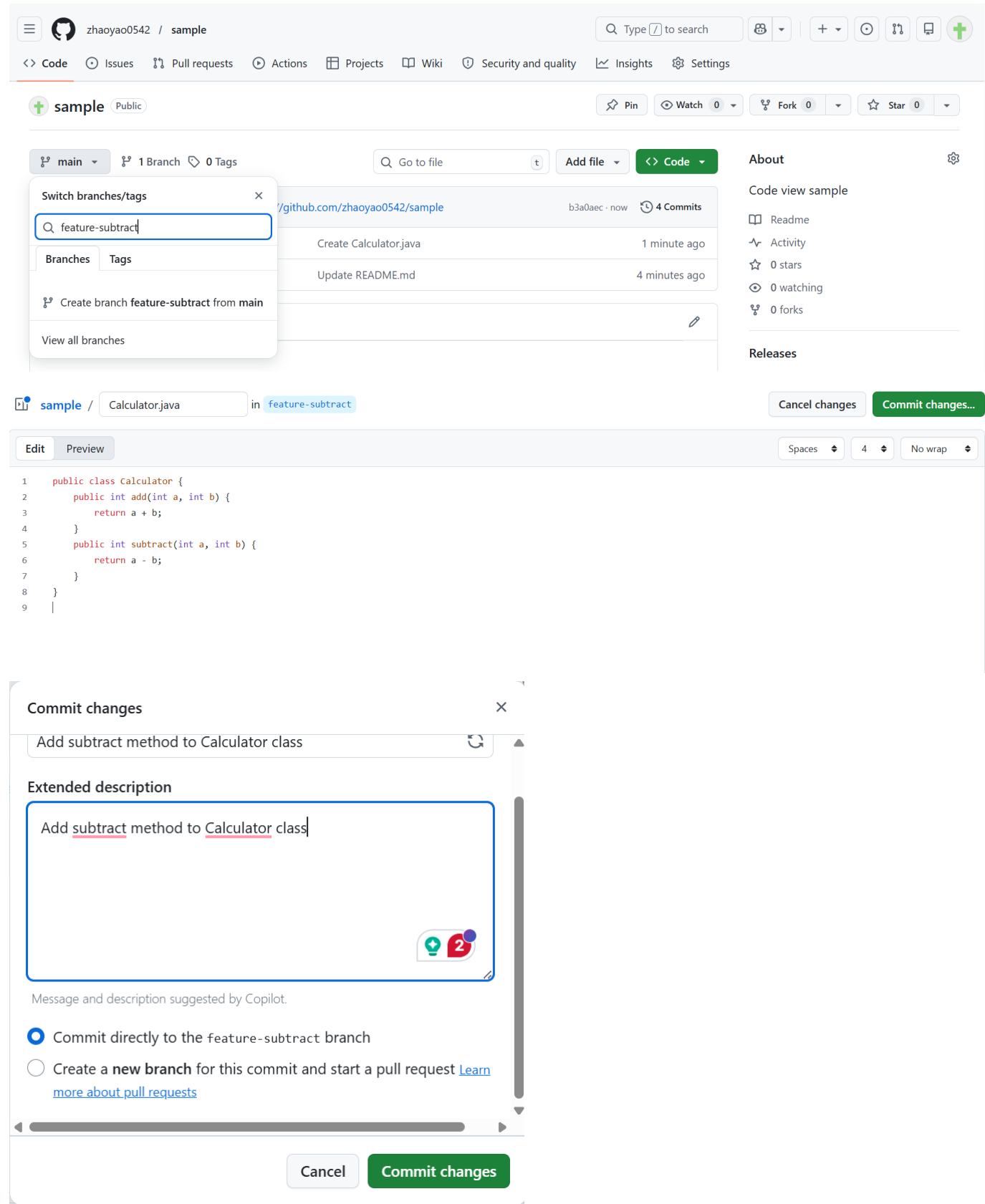
Example Workflow

Assume multiple developers are working on the same project:



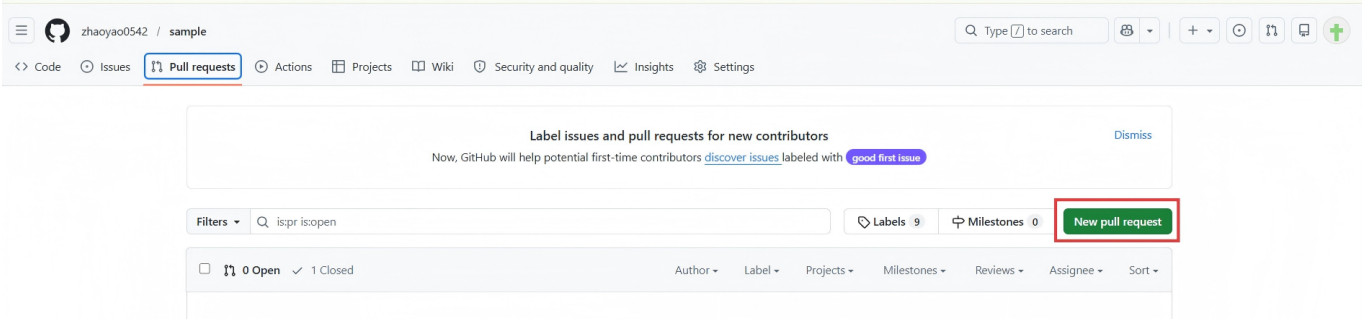
Step 1: Create a New Branch

When you want to make changes, you should create a new branch (e.g., `feature-subtract`) instead of modifying the `main` branch directly:



Step 2: Open a Pull Request

Instead of directly merging your changes to `main`, you can open a pull request:

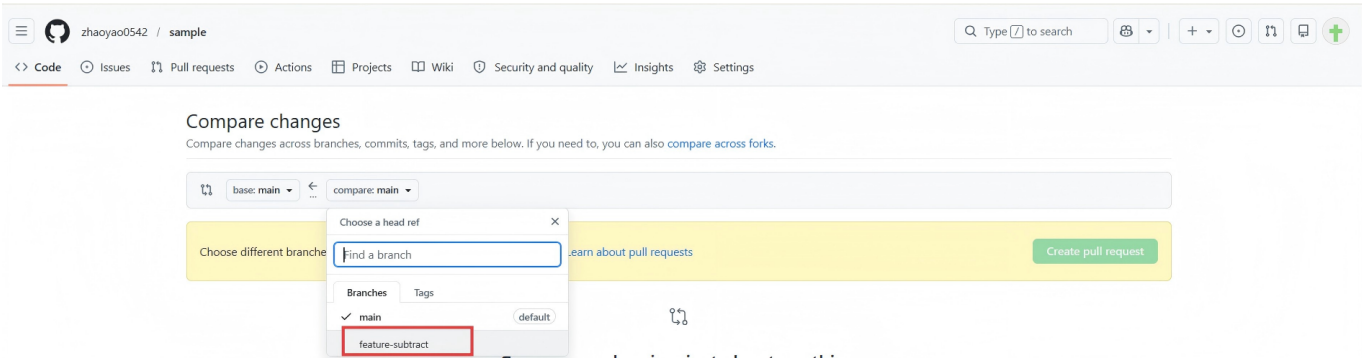


You will need to select:

- **Base branch** (e.g., `main`)
- **Compare branch** (e.g., `feature-subtract`)

This means:

“Request to merge changes from `feature-subtract` into `main`.”



Step 3: Discussion and Review

Your teammates can review your pull request, compare file changes and leave comments:

Checks 0

Files changed 1

Add subtract method to Calculator class

Add subtract method to Calculator class

< Prev

Next >

zhaoyao0542 authored 1 hour ago

Verified

Calculator.java

@@ -2,4 +2,7 @@ public class Calculator {

2 2 public int add(int a, int b) {

3 3 return a + b;

4 4 }

5 - }

5 + public int subtract(int a, int b) {

6 + return a - b;

7 + }

8 + }

Add subtract method to Calculator class #1

Open

 zhaoyao0542 wants to merge 1 commit into main from feature-subtract

Conversation 0

Commits 1

Checks 0

Files changed 1

+

zhaoyao0542 commented 2 minutes ago

Owner ...

Add subtract method to Calculator class

😊

+

Add subtract method to Calculator class

Verified 8bf9519

+

zhaoyao0542 commented now

Owner Author ...

Consider adding JavaDoc for this method.

😊

🔗

✓

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

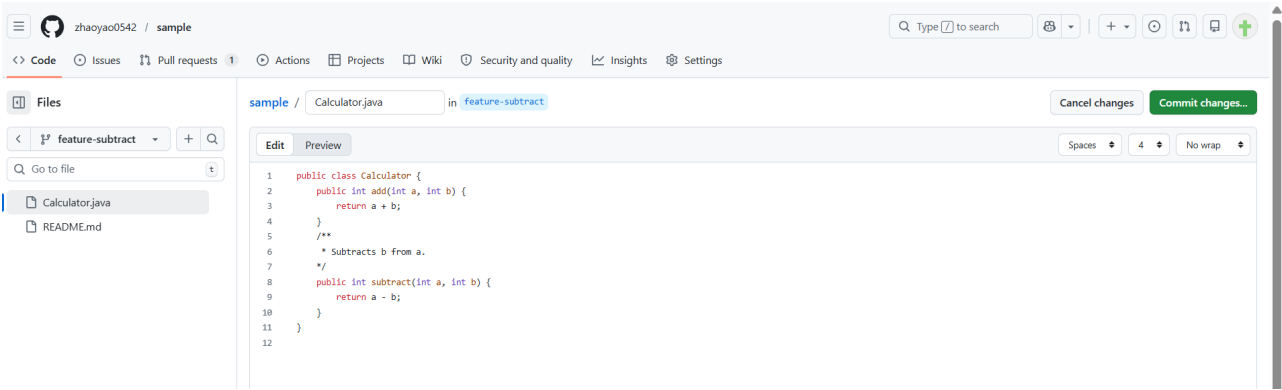
You can also merge this with the command line. [View command line instructions.](#)

Step 4: Update the Pull Request

If reviewers request changes, you can update the pull request by:

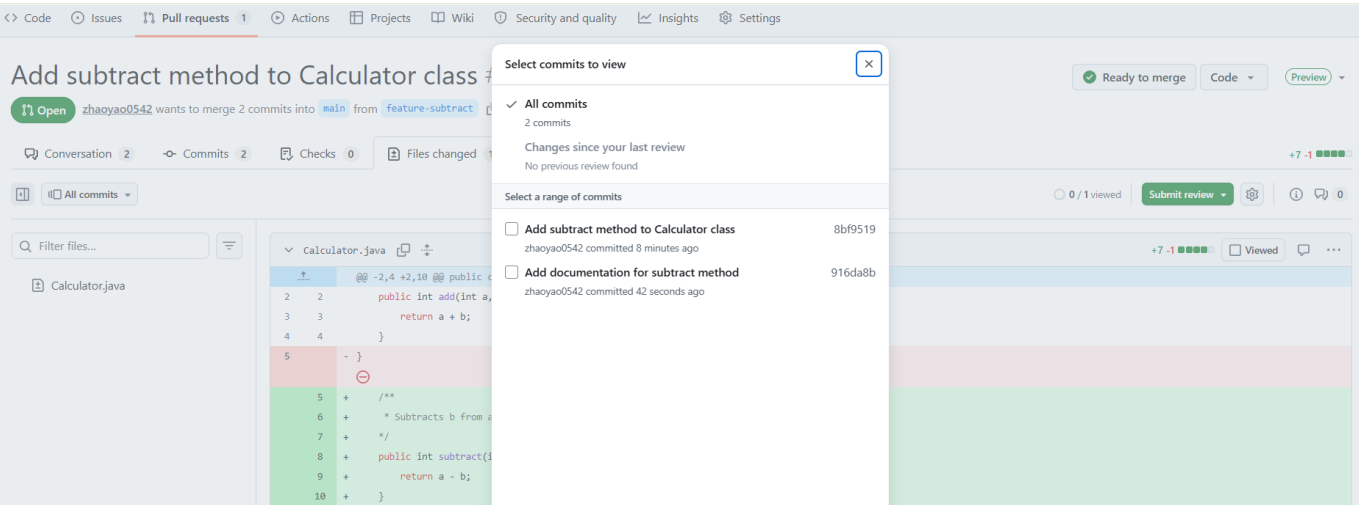
- 1. Making new commits in your branch

2. Pushing the changes



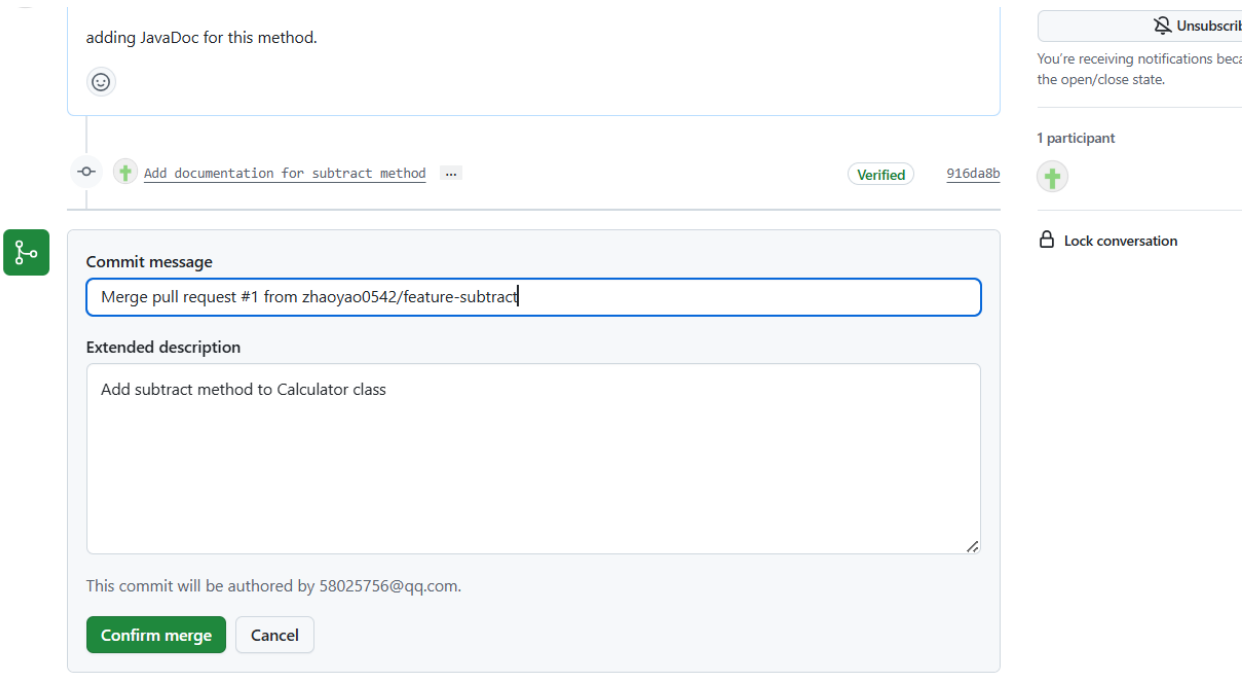
The pull request will update automatically.

The reviewers can check again:



Step 5: Merge or Close

Once the pull request is approved, you can merge it:



Alternatively, if the changes are no longer needed, you can **close** the pull request without merging.

Add subtract method to Calculator class #1

 **Closed** zhaoyao0542 wants to merge 1 commit into `main` from `feature-subtract` 

Conversation 0 Commits 1 Checks 0 Files changed 1

 zhaoyao0542 commented 3 minutes ago Owner ...

Add subtract method to Calculator class



  Add subtract method to Calculator class ... Verified 8bf9519

 zhaoyao0542 commented 2 minutes ago Owner Author ...

Consider adding JavaDoc for this method.



  zhaoyao0542 closed this now

 **Closed with unmerged commits** Delete branch

This pull request is closed, but the `feature-subtract` branch has unmerged commits.

2. Line Comments in Pull Requests

Reviewers can comment on specific lines of code in the **Files changed** tab.

This is useful for:

- Pointing out bugs
- Suggesting improvements
- Asking questions about implementation

Conversation 2Commits 2Checks 0Files changed 1

All commits

Filter files...Calculator.java

Calculator.java

Calculator.java

...

@@ -2,4 +2,10 @@ public class Calculator {

22public int add(int a, int b) {

33return a + b;

44}

55- }

66+ /**

77+ * Subtracts b from a.

88+ */

99+ public int subtract(int a, int b) {

1010+ return a - b;

1111+ }

5+ /**

6+ * Subtracts b from a.

Comment on line R6

zhaoyao0542 now

OwnerAuthor...

*Parameters: a, b

Write a reply

Resolve comment

zhaoyao0542 commented 1 minute ago

View reviewed changes

Calculator.java

44}

55- }

55+ /**

66+ * Subtracts b from a.

No newline at end of file

zhaoyao0542 1 minute ago

OwnerAuthor...

*Parameters: a, b

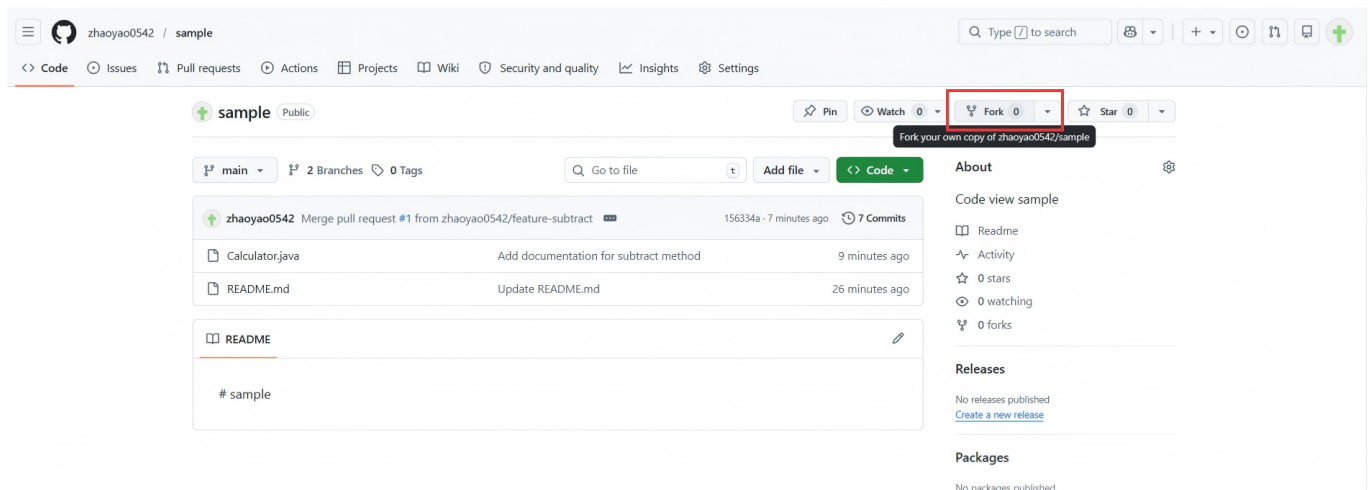
Reply...

Resolve conversation

3. Pull Requests Without Write Access (Fork Workflow)

If you do not have write access to a repository, you cannot push branches directly. Instead, you can use a **fork**.

Step 1: Fork the Repository



This creates a copy of the repository under your own account.

Step 2: Make Changes in Your Fork

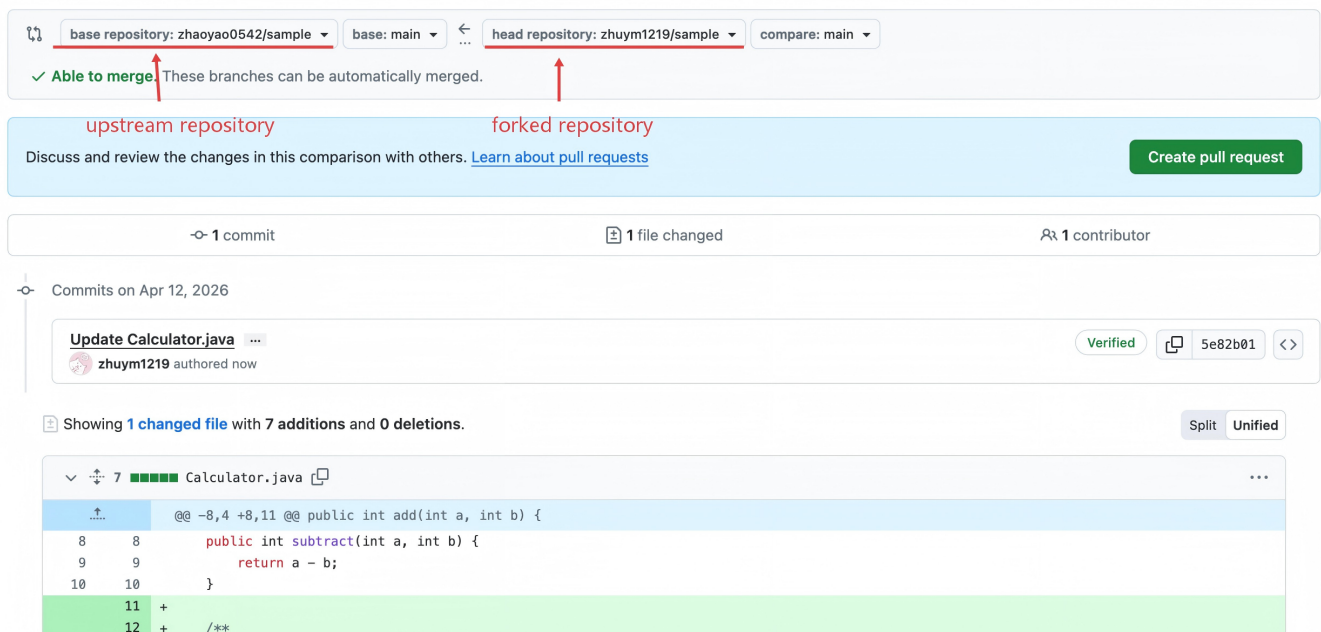
- Modify files
- Commit changes
- Push to your fork

Step 3: Create a Pull Request

You can then open a pull request to the original repository:

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).



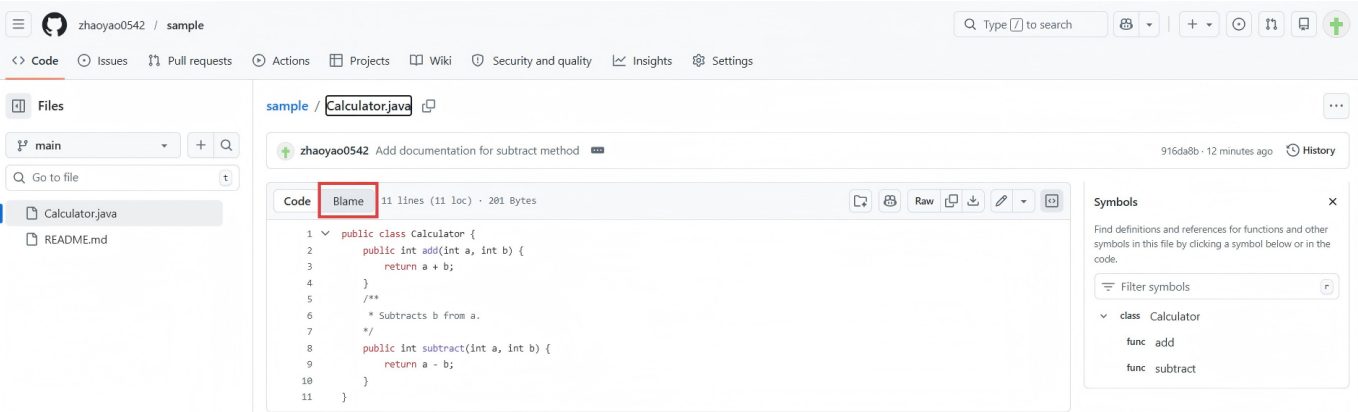
This requests the original repository owner to review and merge your changes.

4. Blame View

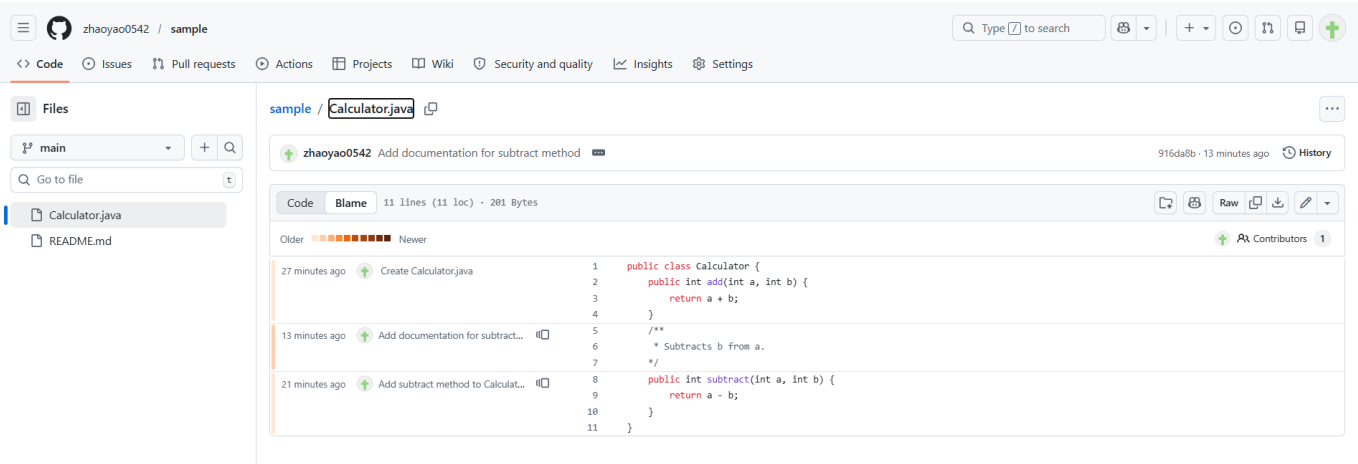
The **Blame** feature allows you to track the history of each line in a file.

Step 1: Open Blame View

Open a file and click **Blame**:



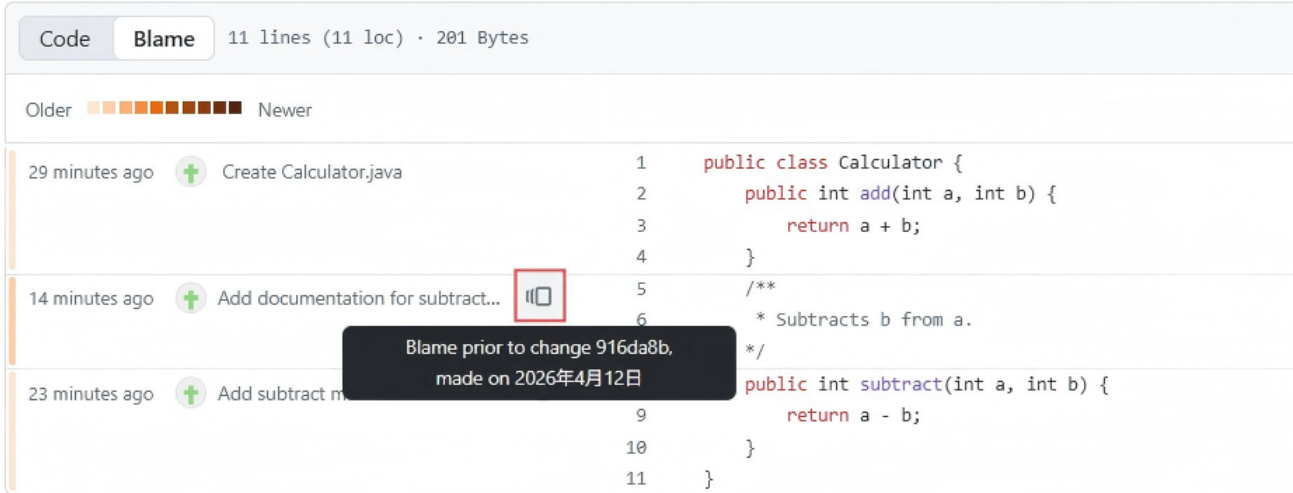
Step 2: Explore Line History



In this view, you can:

- See who last modified each line
- View commit messages
- Navigate to previous versions of the code

Click the highlighted button to trace earlier versions of a specific line.



Code		Blame	8 lines (8 loc) · 159 Bytes
Older ■ ■ ■ ■ ■ ■ ■ ■ Newer			
30 minutes ago		Create Calculator.java	1 public class Calculator { 2 public int add (int a, int b) { 3 return a + b; 4 }
24 minutes ago		Add subtract method to Calculat... 	5 public int subtract (int a, int b) { 6 return a - b; 7 } 8 }

Command Line Alternative

You can also use Git in the terminal:

```
git blame <filename>
```

This shows line-by-line history directly in the command line.

Summary

In this tutorial, you learned:

- How to create and manage pull requests
- How to review and discuss code changes
- How to contribute using forks
- How to track code history using blame

These features are essential for collaborative software development and professional code review workflows.